

Introduction to Exploratory Data Analysis

Course 5. EDA with probability distributions

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Why knowing the distribution matters

- Gaining knowledge or confirming intuition on the data generating process
- Validating data acquisition
- Modeling: as a data fitting tool
- Some methods require that data is distributed in a certain way

Descriptive statistics

Centrality

- Mean
- Mode
- Median

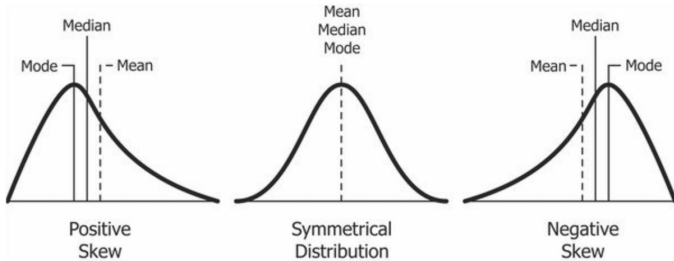
Dispersion

- Standard deviation
- Coefficient of variation
- Percentile

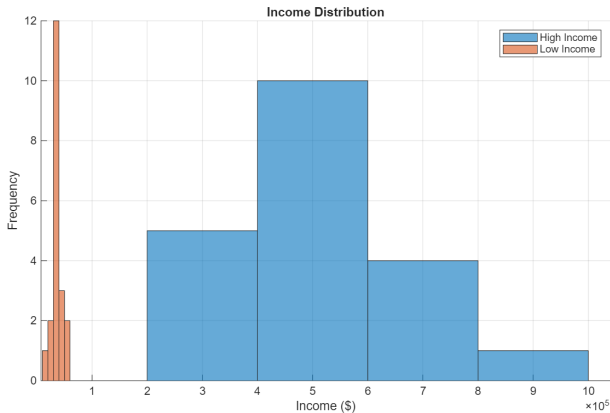
Shape

- Skewness
- Kurtosis

Centrality



Coefficient of variation



$$\mu_{low} = 36780$$

$$\sigma_{low} = 8600$$

$$\mu_{high} = 504356$$

$$\sigma_{high} = 144258$$

Dispersion

Coefficient of variation

$$CV = \frac{\sigma}{\mu} \quad (1)$$

Useful for comparing data with different means.

$$\mu_{low} = 36780$$

$$\sigma_{low} = 8600$$

$$CV_{low} = 0.2679$$

$$\mu_{high} = 504356$$

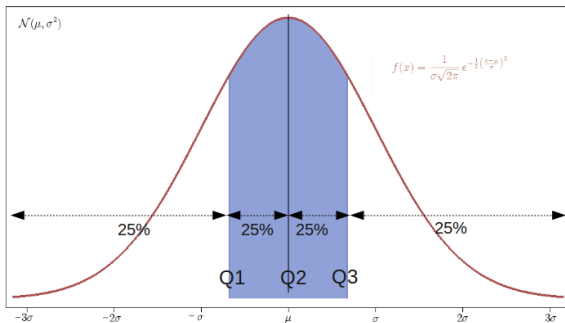
$$\sigma_{high} = 144258$$

$$CV_{high} = 0.1935$$

Percentiles

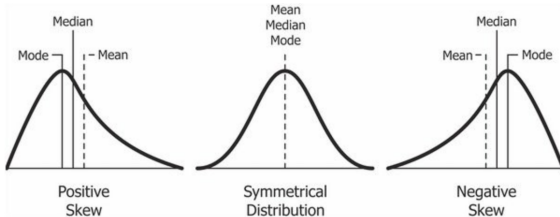
Quantiles divide the data range into intervals having the same probability.

Percentiles are quantiles that divide the data range into 100 groups.



Skewness

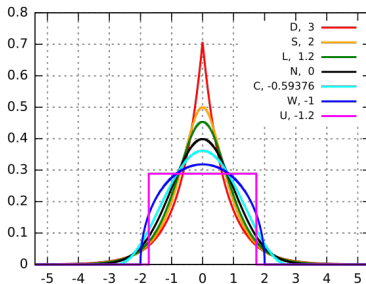
$$Skew(\mathbf{X}) = E \left[\left(\frac{\mathbf{X} - \mu}{\sigma} \right)^3 \right] \quad (2)$$



Measures direction and magnitude of the deviation from the mean.

Kurtosis

$$Kurt(\mathbf{X}) = E \left[\left(\frac{\mathbf{X} - \mu}{\sigma} \right)^4 \right] \quad (3)$$



Measures how much of the probability mass sits in the tails.

Moments

$$Var(\mathbf{X}) = E \left[\left(\frac{\mathbf{X} - \mu}{\sigma} \right)^2 \right] \quad (4)$$

distance from mean, always positive (i.e.) side doesn't matter

$$Skew(\mathbf{X}) = E \left[\left(\frac{\mathbf{X} - \mu}{\sigma} \right)^3 \right] \quad (5)$$

amplifies large distances (power 3) and accounts for sign (asymmetry)

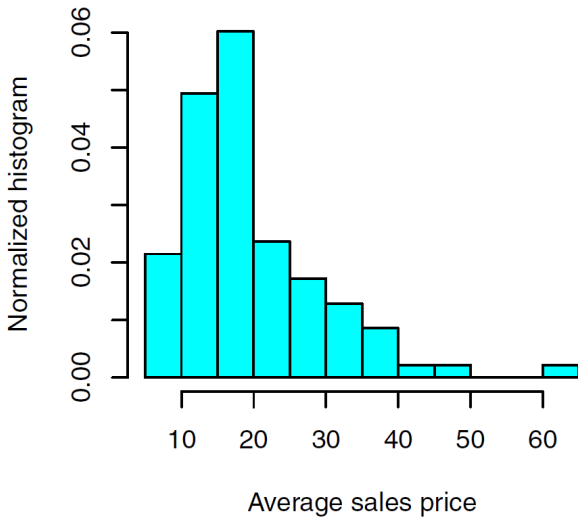
$$Kurt(\mathbf{X}) = E \left[\left(\frac{\mathbf{X} - \mu}{\sigma} \right)^4 \right] \quad (6)$$

further amplifies large distances (power 4), side doesn't matter

For the following examples, try to guess the shape/type of the distribution:

- Prices of new cars in 1993
- Age of mother at the end of pregnancy
- Finish times for male runners in the 2008 NYC Marathon
- City population
- Daily traffic flow

Car prices



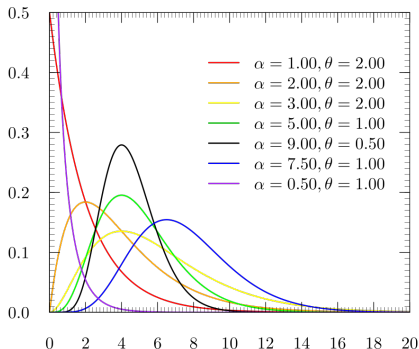
Gamma distribution

Probability density function

$$f(x; \alpha, \lambda) = \frac{x^{\alpha-1} e^{-\lambda x} \lambda^\alpha}{\Gamma(\alpha)}, \text{ for } x > 0, \alpha, \lambda > 0 \quad (7)$$

where $\Gamma(\alpha) = (\alpha - 1)!$

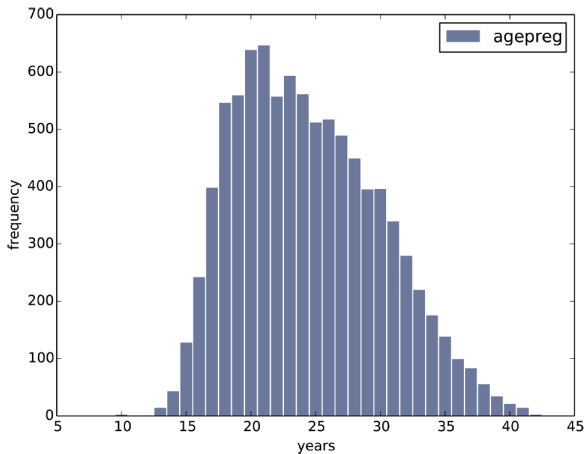
shape parameter, α
rate parameter, λ



Distribution parameters

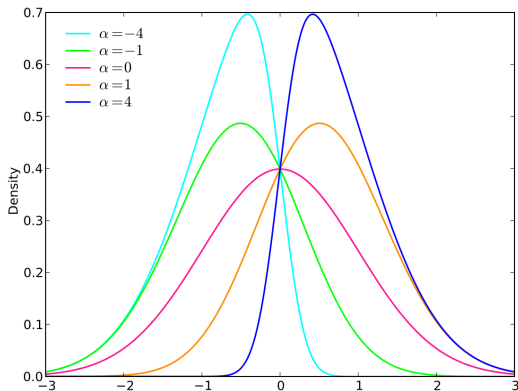
- **Location** parameter, shifts the distribution. $Y = X + \mu$
In the normal distribution, the shape parameter is the mean.
- **Scale** parameter, defines how spread out/concentrated a distribution is. $Y = \sigma X$
In the normal distribution, the shape parameter is the standard deviation.
- **Rate** parameter, inverse of scale parameter.
- **Shape** parameter, any other parameter determining the shape of the distribution.
Skewness and kurtosis can be estimators of the shape parameters in a distribution.

Pregnancy age



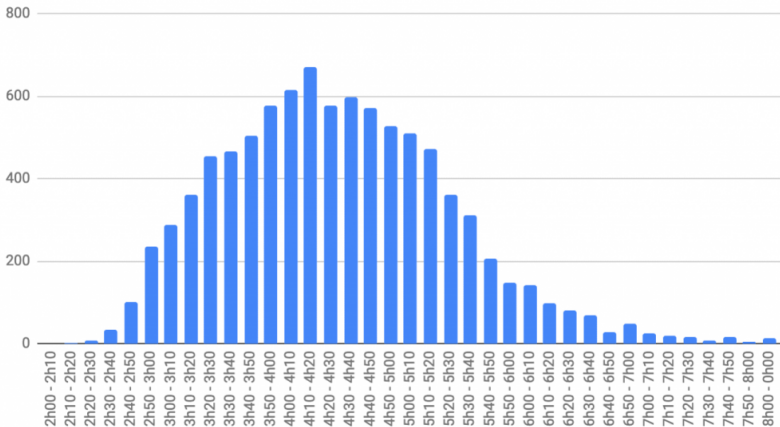
Skew normal distribution

Ex: model IQ, BMI



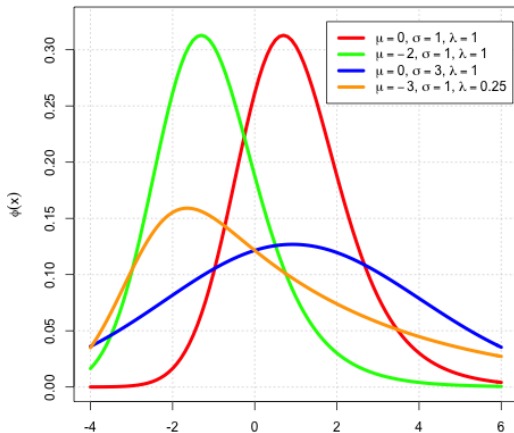
Finish times for male runners in the 2008 NYC Marathon

Male finishers between 35-44 of the 2008 NYC Marathon

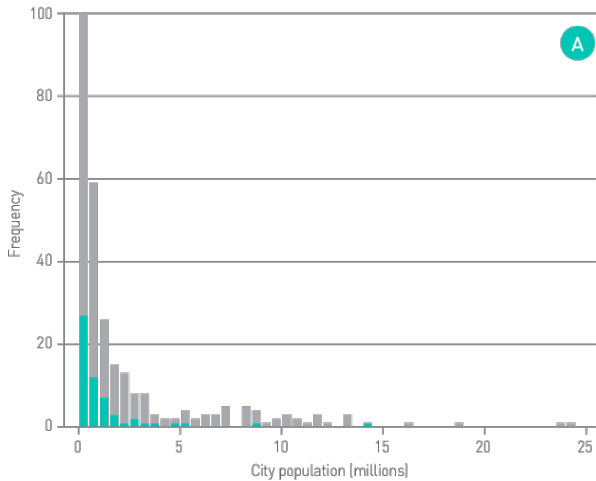


Exponentially modified normal distribution

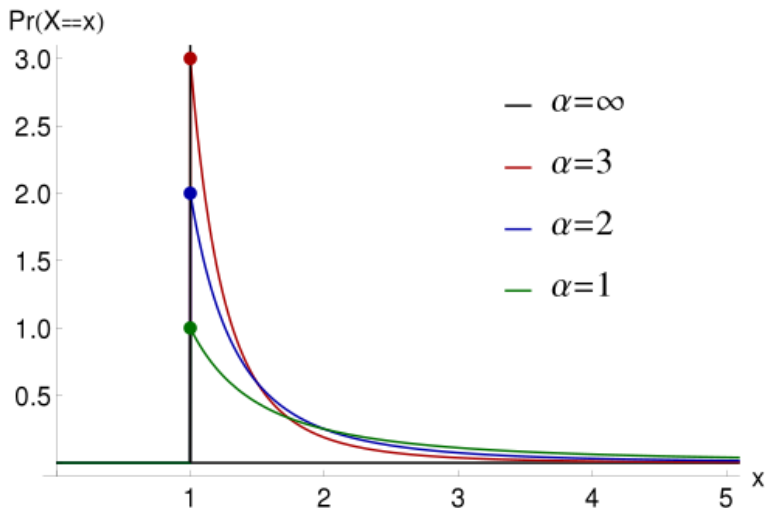
Ex: model reaction times



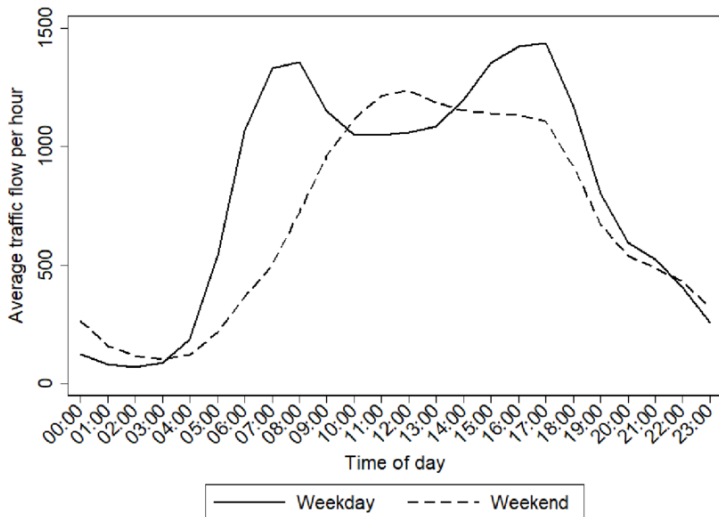
City population



Pareto distribution



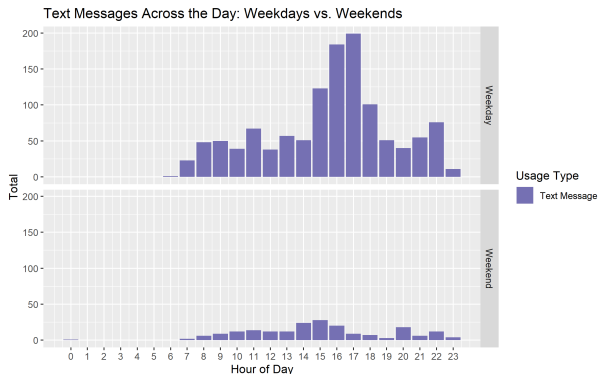
Daily traffic flow



Comparing distributions

Comparing a measured distribution with a theoretical one (fitting)

Comparing two measured distributions (analysis)



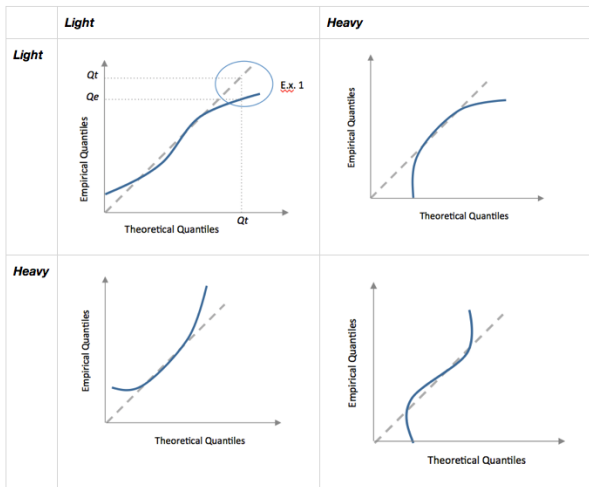
Quantile-Quantile plot

Left Tail (columns)

Right Tail (rows)

Legend

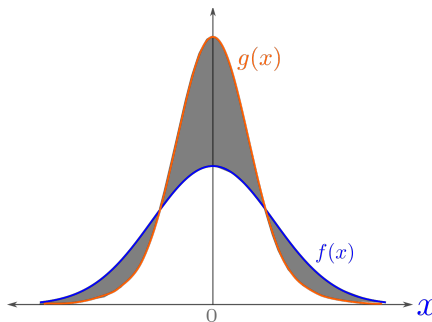
--- $y = x$
— QQ plot



Comparing probability distributions

Total variation

$$\delta(P, Q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)| \quad (8)$$



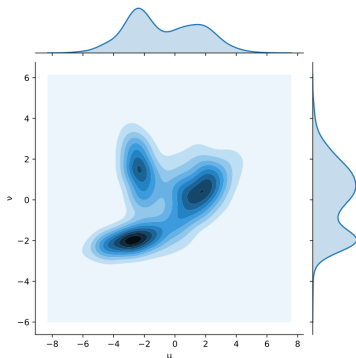
Comparing probability distributions

Wasserstein distance

Minimal cost of transforming (moving) distribution P into distribution Q.

$$W(P, Q) = \inf_{\pi} \left(\frac{1}{n} \sum_{i=1}^n \|P_i - Q_{\pi(i)}\|^2 \right)^{\frac{1}{2}} \quad (9)$$

with π denoting (all) permutations



Encoding information

Information = surprise

$$I(\mathbf{E}) = \frac{1}{\log(p(\mathbf{E}))} \quad (10)$$

for event $\mathbf{E}$

Entropy

- Average level of uncertainty
- Average amount of information needed to describe the variable

$$H(\mathbf{X}) = - \sum_{x \in \mathbf{X}} p(x) \log p(x) \quad (11)$$

Comparing probability distributions

Kullback-Leibler divergence

Measures how much a distribution Q is different from a distribution P .

$$D_{KL}(P||Q) = \sum_{x \in \mathbf{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (12)$$

Interpretation of the KL divergence of P from Q :

Inefficiency (information loss) of using the model (Q) instead of the true distribution (P)

OBS: D_{KL} is not symmetric!